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Performance Evaluation of a Phased-Array Flowmeter for Use in Non-Newtonian Drilling Mud Flow

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Managed pressure drilling (MPD) has become a cornerstone technique in modern drilling operations, enabling precise control of annular pressure and mitigating risks such as kicks, losses, and wellbore instability. A critical element in successful MPD execution is the ability to measure the flow of drilling fluids accurately and reliably under dynamic downhole and surface conditions. Real-time surface flow monitoring provides vital feedback for hydraulic models and control algorithms, but capturing this information accurately, particularly when dealing with complex fluid systems such as oil-based muds (OBM), remains a challenge (Turner and Wilson, 2014; Al-Khafaji et al., 2016).

Traditional flow measurement technologies, while effective in steady-state or single-phase industrial applications, exhibit multiple limitations when applied to non-Newtonian, multiphase, or transient flow regimes. Coriolis mass flowmeters, often regarded as the gold standard in many industries, provide direct mass flow and fluid density measurements. However, they are constrained by pressure limitations typically up to 3,000 psi and are vulnerable to mechanical vibrations, especially in rig environments (Patel and Huang, 2021). Their performance further deteriorates under pulsating flow conditions, in fluids with significant shear sensitivity (Nguyen and Reyes, 2020), or when gas entrainment is present. In addition to entrained gas, Coriolis meters are susceptible to pressure-driven bubble formation (cavitation), which can occur even in nominally single-phase fluids due to sharp pressure drops. These bubbles interfere with the vibration of the flow tube, causing instability and substantial measurement error (Ross and Harvey, 2011).

Differential pressure (DP) devices, such as V-cone and wedge flowmeters, rely on pressure drop across a known restriction and require stable density inputs to infer volumetric flow. These meters tend to drift over time due to erosion or scale buildup on the flow constriction and are susceptible to clogging when exposed to high solids loading, an inherent characteristic of drilling mud. Furthermore, DP meters assume Newtonian flow behavior in their governing equations, limiting their accuracy when used with shear-thinning or thixotropic fluids (Zhao et al., 2022).

Introduction

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Pump stroke counters, commonly used in upstream operations, estimate flow based on pump displacement volume and rotation rate. Their performance suffers over time due to changes in pump efficiency from wear, fluid slippage, or valve degradation. These devices also cannot capture transient variations in flow between pump strokes and do not accurately reflect changes in fluid properties, such as compressibility or temperature-dependent viscosity.

Given these limitations, there is a clear and widespread industry need for a passive, cost-effective, high turndown, pressure-rated, and rheology-agnostic flowmeter capable of reliable performance under harsh drilling conditions and potentially hazardous process chemistry. This study investigates such a solution: a full-bore flowmeter employing a phased-array configuration with internally mounted piezoelectric sensors shielded by a metallic sleeve. The system passively measures flow velocity by analyzing turbulent strain signals along the pipe wall. Volumetric flow is then computed using the known cross-sectional area.

The full-bore phased-array system provides a non-intrusive, pressure-rated solution suited for MPD environments with limited space, high pressure, and abrasive muds. Flow velocity is derived from convective energy patterns identified through frequency-wavenumber spectral analysis of turbulence-induced wall strain. This method passively detects naturally occurring turbulent eddies without requiring intrusive probes or flow restrictions. An embedded digital signal processing (DSP) architecture enables real-time analysis and stable velocity estimation, even under pulsating or transitional flow conditions.

This paper presents a comprehensive evaluation of the phased-array flowmeter in a closed-loop test environment using oil-based muds characterized with non-Newtonian behavior. A Coriolis meter was used as a high-accuracy reference to validate the flowmeter's performance across a broad range of Reynolds (Re) numbers and shear rates.

The study aims to:

i. Characterize the raw sensor response across different flow regimes,
ii. Assess the effectiveness of a Re-based calibration method,
iii. Quantify repeatability and demonstrate robustness under transient operating conditions.

Measurement Principle of the Phased-Array Flowmeter

Sensor Architecture

The flowmeter evaluated in this study is a full-bore, phased-array system equipped with internally mounted piezoelectric sensors permanently bonded to the inner pipe wall and shielded by a metallic sleeve, as illustrated in Figure 1. These sensors are designed to detect dynamic strain fluctuations generated by turbulence in the flowing fluid. Unlike conventional meters relying on mechanical displacement, differential pressure, or active ultrasonics, this system passively detects flow-induced wall strain to infer flow velocity.

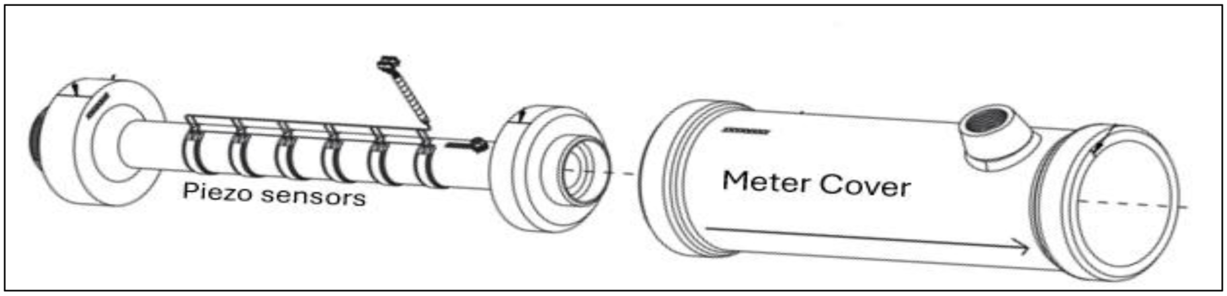

Figure 1—Schematic of the phased-array flowmeter showing internal piezoelectric sensors bonded to the pipe wall and protected by a metallic sleeve.

Micro-Strain Signal Acquisition and Processing

As fluid moves through the pipe, turbulent eddies induce pressure fluctuations along the pipe wall. These are captured as micro-scale dynamic strain signals by the embedded piezoelectric sensors. The signals are digitized and processed in real time using a proprietary algorithm implemented in embedded DSP firmware. The DC component of the strain is effectively rejected by the signal processing, making the velocity signal immune to changes in process pressure or static external loads.

The algorithm performs spectral decomposition and analyzes phase shifts across the sensor array to estimate the convective velocity of turbulent eddies. A key feature of the processing is its use of the frequency–wavenumber domain, where coherent flow eddies appear as energy ridges. The slope of this ridge $\Delta f/\Delta k$, directly corresponds to the convective velocity. where Δf is the frequency difference (Hz) and Δk is the wavenumber difference (rad/m).

The resulting frequency and wavenumber plot (Figure 2) illustrates this principle: energy is concentrated along a diagonal band whose slope increases with flow rate. The vertical ridge is a common mode (DC) signal combined with Speed Of Sound (SOS) signals that typically persist at much smaller wavenumbers. By applying beamforming algorithms and spectral weighting, the system isolates and tracks these peaks even under low signal-to-noise conditions, ensuring robust velocity estimation across laminar, transitional, and turbulent regimes.

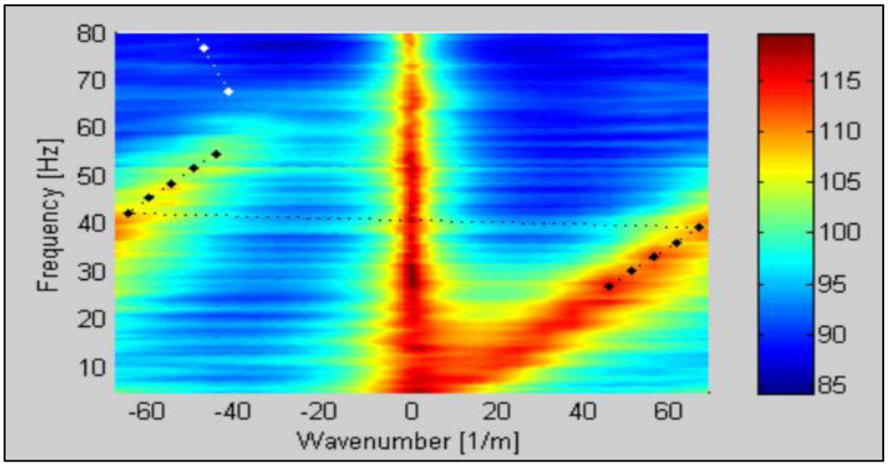

Figure 2—Wavenumber-Frequency plot (k-w) plot showing energy ridge corresponding to downstream flow velocity

Volumetric Flow Calculation

The volumetric flow rate (Q) is calculated from the estimated bulk velocity (V) and the known internal cross-sectional area of the pipe (A), per equation (1).

$$Q = V * A \tag{1}$$

This calculation assumes a fully developed turbulent velocity profile and uniform pipe geometry, conditions satisfied once the flow stabilizes downstream of any mixing or pumping disturbances. To ensure accuracy across a wide range of rheological conditions, the velocity profile estimate is calibrated using a Re-based model. This calibration accounts for the influence of shear-dependent viscosity on the transport of turbulent eddies that generate wall strain, ensuring that the derived velocity reflects the true bulk flow behavior without requiring fluid-specific tuning.

Velocity Calibration Using Re-Based Rheological Modeling

Importance of Rheological Calibration

Drilling fluids, particularly OBMs, often exhibit non-Newtonian behavior, where viscosity varies with shear rate. This rheological property affects the formation and propagation of turbulent eddies in the pipe, which directly influences the wall strain signals detected by the phased-array sensors (Growcock et al., 1994; Kelessidis et al., 2006; Zamora and Roy, 2016). Since the system derives flow velocity from the movement of these turbulence-induced pressure fluctuations, the meter's response is inherently sensitive to how rheology shapes their development and behavior.

Assuming Newtonian behavior across all flow conditions introduces a systematic bias, particularly in shear-thinning regimes where apparent viscosity decreases with increasing flow rate. In these cases, the velocity field becomes more plug-like, which reduces wall strain signal strength and can lead to underestimation of flow velocity (Chhabra and Richardson, 2008). In contrast, shear-thickening (dilatant) fluids may produce stronger wall gradients, potentially leading to velocity overestimation if not properly accounted for.

To address this, a Re-number-based calibration strategy is employed. This method enables a robust and transferable velocity estimation framework across a wide spectrum of drilling fluid types without requiring fluid-specific recalibration. Rather than assuming a fixed viscosity, Re dynamically reflects the evolving rheological state of the fluid, allowing calibration to adapt to different flow regimes from laminar to fully turbulent.

This study focuses on steady-state shear-thinning fluids, typical of field-grade OBMs. It does not attempt to compensate for more complex time-dependent behaviors, such as thixotropy or rheopexy, which are generally minimal in well-conditioned OBMs under stable flow conditions (Zamora and Roy, 2016).

Estimating Shear Rate for Non-Newtonian Flow

To enable Re-based calibration, an effective shear rate ($\dot{\gamma}$) must first be estimated. Two models are used depending on the desired level of fidelity:

a. **API 13D Method** For fully-developed laminar flow of power-law fluids in a cylindrical pipe, the API 13D provides the apparent shear rate:

$$\dot{\gamma} = \frac{8V}{D} \tag{2}$$

where V is the axial velocity and D is the pipe internal diameter (API RP 13D, 2022).

b. **Weissenberg–Rabinowitsch–Mooney (WRM):** To refine this further, particularly for fluids with measurable shear-thinning behavior, the WRM equation adjusts the wall shear rate based on the shape of the velocity profile:

$$\dot{\gamma} w = \left(\frac{3n' - 1}{4n'}\right) * \frac{8V}{D} \tag{3}$$

where n' is the flow behavior index derived from rheometric data (Mohammadi et al., 2012).

Herschel–Bulkley Model for Effective Viscosity

Using the estimated shear rate, the fluid's effective viscosity (μ_{eff}) is computed via the Herschel–Bulkley model, a widely accepted rheological equation for OBMs:

$$\mu_{eff} = \frac{\tau_y}{\dot{\gamma}} + K \cdot \dot{\gamma}^{n-1} \tag{4}$$

Where:

- τ_y: yield stress (Pa),
- K: consistency index ($Pa \cdot s^n$),
- n : flow behavior index (dimensionless),
- $\dot{\gamma}$: estimated shear rate (s^{-1}).

This equation accounts for both the initial resistance to flow (yield stress) and the non-linear viscosity response to increased shear, which are common in field muds. Rheological parameters τ_y, K, and n are experimentally determined using a rotational viscometer and fitted via regression (Zamora and Roy, 2016; Hemphill et al., 1993).

Re and Calibration Function

Using the effective viscosity (μ_{eff}), Re is defined as (Al-Khafaji et al., 2016; Zamora and Roy, 2016) :

$$Re = \frac{\rho V D}{\mu_{eff}} \tag{5}$$

Where:

- σ: fluid density (kg/m^3).
- V: uncorrected velocity estimated from spectral analysis (m/s).
- D: internal diameter of the pipe (m).

- μ_{eff}: effective dynamic viscosity (Pa·s).

To account for rheology-induced deviations in measured velocity, an empirical calibration function *f(Re)* is applied:

$$V_{corrected} = V_{measured} \cdot f(Re) \tag{6}$$

The function *f(Re)* is derived from experimental correlation against Coriolis reference data and reflects how flow development and turbulence strength vary with Re:

- At **Re > 10^4^**, $f(Re) \approx 1.0$, indicating a fully turbulent velocity profile and negligible rheology influence.
- At **Re < 2000:** increased deviation due to damped turbulence and lower wall strain energy.
- In the **transitional range** *f*(*Re*) follows a monotonic trend as turbulence strength increases.

Rather than serving as a per-fluid correction, this function is a **physics-informed** calibration that maps the strain-derived velocity to a reference standard while inherently accounting for how viscosity and flow regime affect turbulent eddies propagation and dissipation.

Practical Implementation

The Re-based calibration methodology is designed for seamless integration into real-time flow monitoring systems, particularly in environments where mud properties change dynamically (Patel and Huang, 2021). Its implementation enables automated compensation for rheological variation without interrupting operations.

- The Re-based calibration is applied in real time and within the flowmeter`s firmware.
- Rheological parameters may be loaded as part of the drilling fluid program or inferred via lab testing.
- The calibration curve is continuously applied to account for changes in fluid properties over time (e.g., due to cuttings, temperature, or additive concentration).

This framework ensures that flow velocity measurements remain accurate under evolving field conditions, without requiring frequent manual adjustments or per-fluid recalibration.

Performance Evaluation and Validation Using OBM Flow Loop Testing

Test Overview

To evaluate the performance of the phased-array flowmeter under realistic drilling conditions, a comprehensive flow loop testing was conducted using OBMs exhibiting shear-thinning behavior. The objective was to validate sensor response across a wide range of Re and flow regimes, and to benchmark the velocity output against a calibrated Coriolis reference meter.

Testing focused on the following performance criteria:

- Accuracy of velocity estimation under varying flow and rheology.
- Consistency and repeatability of signal acquisition under stable and transient operation.
- Robustness against pulsation, entrained gas, and mud conditioning effects.

The evaluation was conducted using an in-house mud flow loop capable of simulating field conditions and leveraged the Re-based calibration methodology described in Section 3 to account for rheological effects during testing.

Test Loop Configuration

The in-house mud flow loop is schematically shown in Figure 3. It consists of three main sections: a mud tank skid, dual pump skids (500 HP and 1600 HP), and a metering section where the reference Coriolis and phased-array flowmeters are installed. Key loop components included:

- **500 HP and 1600 HP pump skids**: Used alternately or in parallel to reach desired flow rates and pressure conditions. Each skid featured step-down piping and check valves to regulate turbulent flow and flow direction.
- **Mud tank skid**: Equipped with agitation and mixing control to ensure homogeneity of the OBM during recirculation. This prevented the settling of weighting agents and minimized rheological drift.
- **Measurement section**: The 4-inch stainless steel pipe segment where both meters were installed. The Coriolis meter (reference) was positioned upstream of the phased-array meter to ensure that reference data was unaffected by downstream disturbances. A pressure dampener and backpressure choke were installed downstream to simulate typical backpressure seen in MPD systems.
- **Instrumentation**: High-resolution pressure transducers, temperature probes, and density meters were installed upstream and downstream of the test section. Data were logged at 10 Hz resolution using a dedicated data acquisition system.

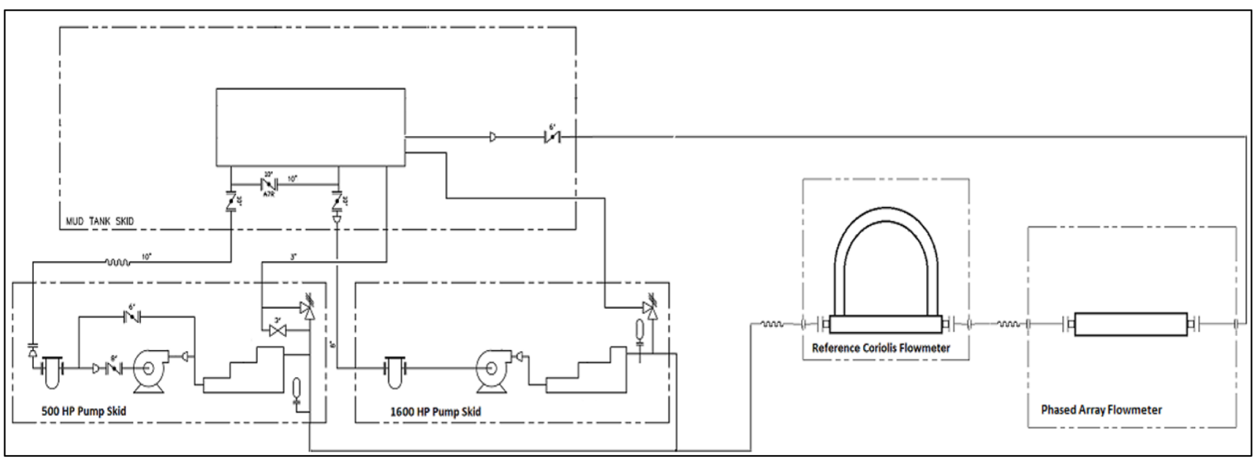

Figure 3—Schematic of OBM Recirculation Flow Loop with Reference and Phased-Array Flowmeters

Flow conditioning was achieved via inline static mixers installed before the measurement section to ensure a fully developed velocity profile. The meter installation adhered to best practices with at least 10D upstream and 5D downstream straight pipe lengths, consistent with ISO 5167-1 requirements.

The phased-array meter consisted of a full-bore pipe with piezoelectric sensors mounted internally beneath a welded protective sleeve. The electrical leads were routed through a sealed bulkhead and terminated at a local data acquisition enclosure with analog-to-digital conversion and signal conditioning circuits.

Fluid Rheology Characterization

Viscosity measurements were performed using a Fann 35 viscometer in accordance with API 13B-2 procedures. Multiple mud batches were prepared, and their rheological parameters were fit to the Herschel–Bulkley model. Table 1 shows the derived parameters:

Table 1—Rheological Parameters for OBM Batches Used in Calibration

Batch	Yield Stress (Pa)	Consistency Index ($Pa \cdot s^n$)	Flow Behavior Index (n)
OBM-A	3.2	0.45	0.76
OBM-B	2.8	0.52	0.70

These parameters were used to compute effective viscosity and Re at each test point, serving as the basis for applying the Re-based calibration function discussed in Section 3.

Data Collection Protocol

At each setpoint:

- Flow was stabilized for 2-3 minutes.
- Phased array data were collected continuously for 60 seconds.
- Reference Coriolis measurements and fluid density were recorded.
- Measurements were repeated to establish repeatability.

Velocity was estimated using the spectral slope technique and then calibrated via the Re-dependent function, f(Re) (Equation 6), as derived from OBM-specific test data. The calibrated velocity was subsequently compared to the Coriolis reference to evaluate accuracy.

Results and Discussion

Spectral Velocity Estimation

Figure 4 presents the frequency wavenumber (f-k) power distribution at a representative flow rate of 127 m^3/h. The high-energy ridge sloping toward the positive wavenumber axis corresponds to downstream-propagating turbulent eddies. The slope of this ridge, $\Delta f/\Delta k$, determines the convective velocity. A secondary, less intense ridge in the negative wavenumber quadrant indicates upstream reflections, which were excluded from velocity calculation.

The coherent structure and high signal-to-noise ratio (SNR) of the dominant ridge confirm the sensor's ability to resolve convective flow velocity using passive strain measurements without external excitation. The f-k approach proved particularly robust during pulsating and unsteady flows, where time-domain methods are more prone to noise.

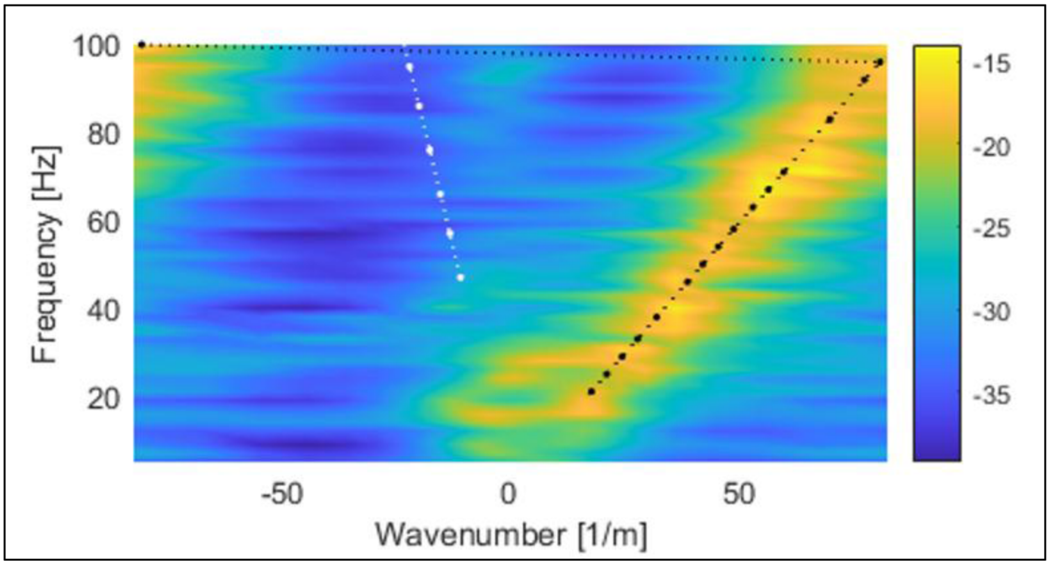

Figure 4—Frequency–Wavenumber (f–k) Spectrum Showing Convective Velocity Ridge

Calibration Accuracy and Re Dependence

Initial velocity estimates from raw strain signals showed increasing accuracy as flow entered the turbulent regime. At lower Re where shear rates are insufficient to generate strong, coherent turbulence, the system under-responded due to reduced wall strain signal strength. These deviations are not sensor artifacts but reflect turbulence-based measurement limitations in laminar and transitional flows.

As shown in Figure 5, pre-calibration velocity estimates exhibited a clear dependence on Re:

- At **Re < 2000** (laminar regime), errors exceeded **20%** due to low convective energy available for detection.
- In the **2000–8000** transitional regime, velocity errors dropped below **10%**, reflecting partial turbulence development.
- At **Re > 10,000**, when the flow is fully turbulent, velocity errors converged to within **±3%**, showing high measurement stability.

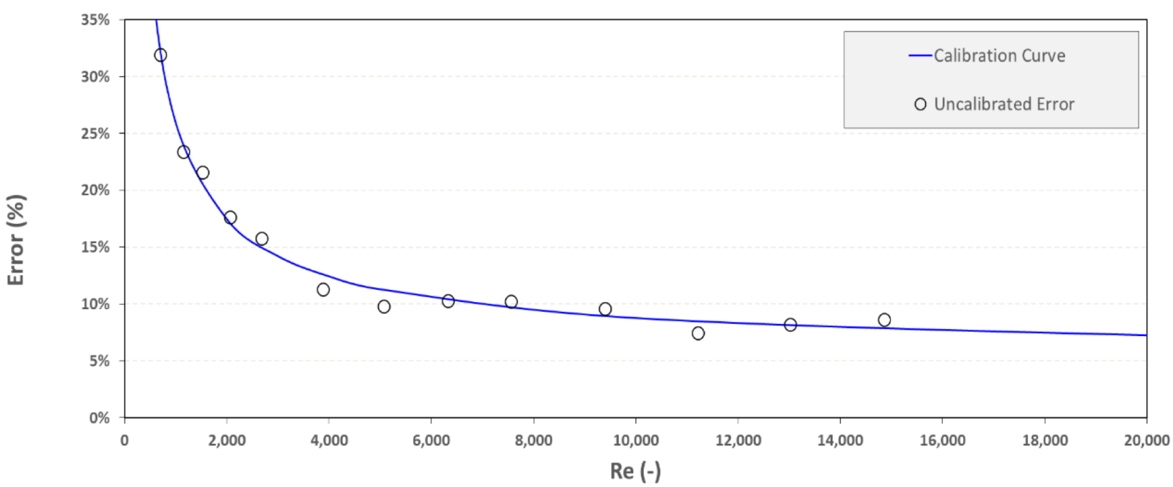

Figure 5—Pre-Calibration Velocity Error vs. Re and Corresponding Empirical Fit

To account for these trends, a Re-based calibration model, as defined in Section 3, was applied. Figure 6 shows the effect of this model: post-calibration velocity estimates are tightly clustered within ±5% of the 1:1 reference line, including in transitional conditions.

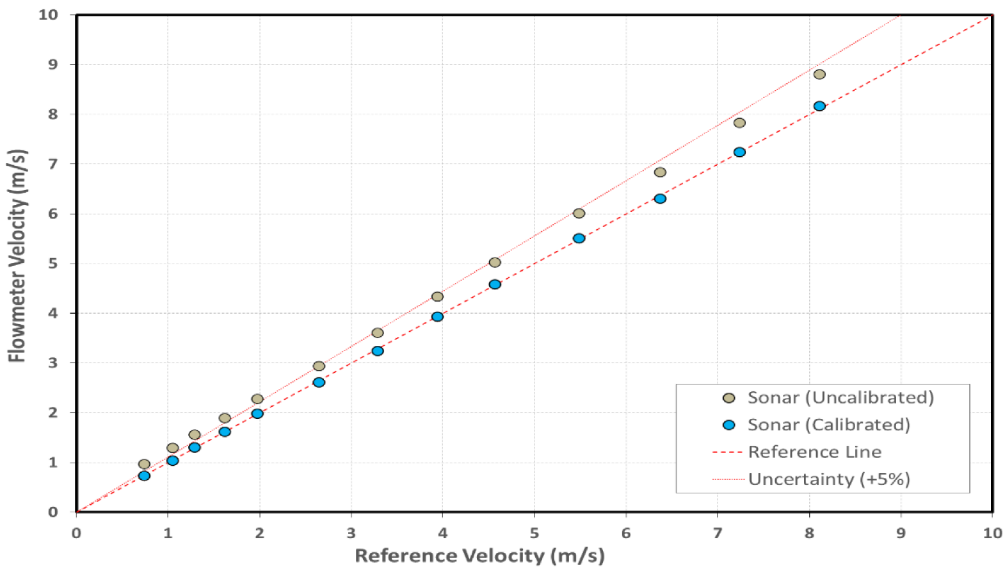

Figure 6—Calibrated and Uncalibrated Velocity Versus Reference: Impact of Re-Based Model

This calibration framework is a physically informed transformation based on non-Newtonian fluid behavior. Rather than correcting for instrument error, it accounts for the relationship between flow regime and signal strength, ensuring reliable velocity output across varying rheology without needing per-fluid recalibration.

Summary of Calibration Trends

The test matrix spanned a wide range of Re and flow conditions. Calibration findings include:

- **Minimum calibration factor**: 0.052 (at Re < 2000)
- **Maximum calibration factor**: 0.639 (transitional regime)
- **Mean factor across matrix**: ~0.0967
- **Post-calibration standard deviation**: 1.18%

Figure 7 illustrates the empirical calibration curve f(Re), illustrating that lower Re corresponding to laminar or transitional flow require greater adjustment due to reduced wall strain signal strength. As turbulence develops, the calibration factor converges asymptotically toward a stable minimum, indicating that minimal adjustment is needed in fully developed turbulent flow.

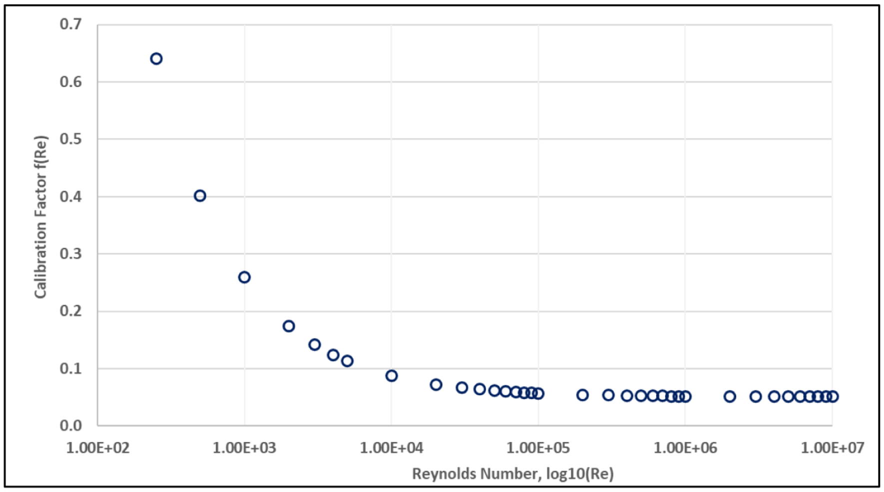

Figure 7—Calibration factor f(Re) versus Re (log scale)

This trend was further quantified by binning the data into distinct Re ranges. Table 2 presents the resulting statistics, offering a clear view of how calibration performance evolves with flow regime.

Table 2—Velocity Error and Calibration Factor by Re

Re Range	Mean Error (%)	Std. Dev. (%)	Calibration Factor Range
< 2000	+18.7	3.9	0.51 – 0.64
2000–8000	+9.4	2.2	0.14 – 0.50
> 8000	+2.1	1.1	0.05 – 0.13

Flow Rate Accuracy and Repeatability

Volumetric flow was calculated using the Re calibrated velocity to the known pipe cross-sectional area. Post-calibration flow rates showed excellent agreement with Coriolis reference values:

- Mean flow rate error: −0.0004%
- Standard deviation: ±0.78%
- 95% confidence interval: ±0.42%
- Maximum absolute deviation: < ±1.31%

The system exhibited repeatability within ±1.2% across multiple trials conducted under consistent flow conditions and varying OBM shear histories. No signal drift or recalibration was required throughout testing.

These findings validate the stability of the spectral processing algorithm and confirm the viability of passive strain-based high-accuracy flow velocity measurement in dynamic drilling environments.

Overall Performance Summary

The phased-array flowmeter achieved reliable and high-accuracy performance under a wide range of oil-based mud conditions, confirming its suitability for field deployment in MPD operations. Key highlights include:

- **Flow rate accuracy**: ±1.0% (95% CI) post-calibration
- **Velocity accuracy**: ±3.0% across full Re range

- **Refresh Rate**: <1 second with real-time output
- **Design benefits**: Non-intrusive, no moving parts, zero pressure drop
- **Operational robustness**: Maintained performance after >60 hours of OBM exposure without recalibration

These results demonstrate the meter's robustness against rheological variability, pulsation, and dynamic shear conditions typical of demanding drilling environments.

Conclusion

Surface flow measurement in OBM remains a critical gap in MPD. This study presents a field-deployable, non-intrusive phased-array flowmeter that offers accurate, real-time flow rate and velocity measurement in OBM environments without pressure drop or flow restrictions.

Key outcomes include:

- **High Accuracy in Complex Fluids**: Post-calibration flow rate error was consistently maintained within ±1.0% (95% CI) across the full Re range.
- **Rheology-Responsive Calibration**: A Re-based calibration model compensated for non-Newtonian effects, enabling reliable performance in both laminar and transitional regimes.
- **Robustness Under Dynamic Conditions**: The passive spectral estimation algorithm remained stable under pulsation, entrained air, and variable shear histories. Repeatability remained within ±1.2% over multiple days of testing.
- **Non-Intrusive and Durable**: The system, with no moving parts or flow obstructions, operated for over 60 hours in high-solids OBM without signal drift or degradation.

Beyond validating the sensor performance, this work introduces a rheology-aware calibration methodology that eliminates the need for per-fluid tuning. This enables adaptive flow monitoring without manual intervention critical for real-time MPD operations.

In summary, the phased-array flowmeter offers a step-change in surface flow diagnostics for MPD, Underbalanced Drilling, and other multiphase applications. Future work will focus on deployment in live wells and extension to a broader range of mud types and geometries.

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